

Introduction to Social Data Analytics:

Lecture 2 Part 1

Today

- ▶ Introduce empirical module: Opportunity Insights data
- ▶ Introduce Stata
 - ▶ graphical user interface
 - ▶ Do-files
 - ▶ Basic commands for understanding your dataset

Intergenerational Mobility

- ▶ If you are born from low-income parents, what is the chance you move up the income distribution?
- ▶ Higher education is often viewed as a path for upward mobility
 - ▶ Differences in quality across colleges and rising costs of college may mean lower access/success for lower-income students
- ▶ Today we will introduce data made publicly available by **Opportunity Insights** that allows us to understand variation across colleges in the extent that they promote intergenerational mobility

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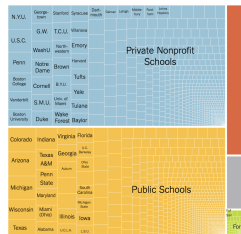
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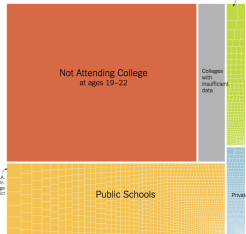
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Top 1%



Bottom 20%



Opportunity Insights Data

- ▶ The data comes from federal income tax returns
- ▶ Researchers at Opportunity Insights have linked to federal tax returns to tax records of universities and data from Pell grants
- ▶ Result is a dataset with:
 - ▶ Children linked to their parents (children are reported as dependents)

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- ▶ Result is a dataset with:
 - ▶ Children linked to their parents (children are reported as dependents)
 - ▶ Income data for both children and parents
 - ▶ Children linked to the universities they attended
- ▶ This was a **huge** data task that gives us a new way to study an important question: how do colleges vary in the extent that they promote intergenerational mobility

Intergenerational Mobility by College

- ▶ To measure intergenerational mobility, we need to compare parent's income to income of their children
- ▶ Inflation (general rise in prices) will make this difficult to do by looking directly at income
- ▶ To measure intergenerational mobility, we will focus on **quantiles**

Quantiles

- ▶ Imagine we have 10 individuals and we sort them based on income (in the thousands per year)

$\{5, 6, 20, 29, 38, 42, 55, 62, 80, 88\}$

- ▶ We can break up the data into equal-sized groups based on income
- ▶ Each equal-sized group is a separate quantile

Example (Quintiles)

- Imagine we break the 10 individuals into 5 groups (or 5 quantiles)

$$\underbrace{\{5, 6\}}_{Q1}, \underbrace{\{20, 29\}}_{Q2}, \underbrace{\{38, 42\}}_{Q3}, \underbrace{\{55, 62\}}_{Q4}, \underbrace{\{80, 88\}}_{Q5}$$

- When we break up the observations into 5 equal-sized groups, this is referred to as breaking the data into **quintiles**
- Q1 is referred to as the first quintile (or lowest quintile)
- Q5 is referred to as the fifth quintile (or highest quintile)

Quintiles and Intergenerational mobility

- ▶ To measure intergenerational mobility, we need to compute the fraction of individuals coming from parents that have **low income** that are able to achieve **high income**
- ▶ The researchers at Opportunity Insights use quintiles to define **low income** and **high income**
- ▶ **Low income:** individuals with parents in the bottom quintile of earnings (Q1)
- ▶ **High income:** individuals that reach the top quintile of earnings (Q5)

College Mobility Rate

- ▶ Mobility rate for a college defined as

$$\text{mobility rate} = \underbrace{(\text{frac. of students from Q1})}_{\text{Access}} \times \underbrace{(\text{frac. of Q1 that reach Q5})}_{\text{Success}}$$

- ▶ Intuition: a college that admits a large fraction of low-income students (first term) and these students become high earners (second term) will promote intergenerational mobility
- ▶ This is the measure of mobility utilized by researchers at Opportunity Insights, but there are alternative measures that could be constructed

Getting Specific

- ▶ How do we (exactly) determine if a student comes from a low-income background?
- ▶ Take all parents that have a child in the same year (e.g. 1985)
- ▶ Compute average income for these parents when children are 15-19 (i.e. from 2000-2004)
- ▶ If parents' income is in bottom 20 percent over this period (i.e. in Q1), then classify students as coming from low-income background

Getting Specific

- ▶ How do we (exactly) determine if a student becomes a high earner?
- ▶ Take all individuals that are the same age and calculate earnings in 2014
- ▶ If individual's income is in the top 20 percent relative to individuals the same age (i.e. in Q5), then classify student as a high earner
- ▶ Now that we understand the variables we will be exploring, let's learn how to use Stata!

The Stata Graphical User Interface (GUI)

The screenshot displays the Stata/SE 16.1 graphical user interface. The top menu bar includes options like Open, Save, Log, Viewer, Graph, Do-file Editor, and Data Editor. The main window is divided into three panes:

- Results Window:** Displays the Stata logo, version 16.1, and the text "Statistics/Data analysis". It also shows the "Special Edition" logo and copyright information for StataCorp. The Stata license information is visible: "Stata license: Unlimited-user network, expiring 31 Jul 2021" and "Serial number: 401609205573".
- Variables Window:** A table with two columns: "Name" and "Label". It is currently empty.
- Properties Window:** A table with two columns: "Name" and "Value". It is currently empty.

The Command window is at the bottom, showing the command prompt.

Results Window

Variables Window

Properties Window

Command Window

Stata/SE 16.1

Open Save Log Viewer Graph Do-file Editor Data Editor

More Break Search

Results

Stata logo

16.1

Statistics/Data analysis

Special Edition

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4905 Lakeway Drive
College Station, Texas 77845
800-STATA-PC <https://www.stata.com>
979-696-4600 stata@stata.com
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Serial number: 401609205573

Command

Variables

Name	Label
------	-------

Properties

Name	Value
Frame	default
Filename	
Label	
Notes	
Variables	0
Observations	0
Size	0
Memory	64M
Sorted by	

Variables

Properties

Frame default

Filename

Label

Notes

Variables 0

Observations 0

Size 0

Memory 64M

Sorted by

Users/Dave/Dropbox/Teaching/EP5/online/behavioral/llns/code

Command Window

- ▶ You can type in code through the command window
- ▶ Pressing enter will execute (or run) the code
- ▶ When we learn how to use Stata as a calculator, this is where we will type the code

Results Window

- ▶ When we execute code, any resulting output will be displayed in the Results window
- ▶ When we learn how to use Stata as a calculator, this is where the results will be displayed

Variables Window

- ▶ When we load data, the name of the variables in the dataset will appear in the first column
- ▶ Many times, datasets come with labels, that give us more description about the variable
- ▶ For example, in the figure below there is one variable named `income` with the label `Total Annual Income`

[illegible]

Properties Window

- ▶ This window gives us more information about the variables and data loaded in memory
- ▶ For example, in addition to the name and label of a variable, the type (i.e. numeric or string) and format of the variable (i.e. how the variable will be displayed)
- ▶ Additionally there is information about the size of the data (number of variables, observations, etc.)

Using Stata as a calculator

- ▶ To use Stata as a calculator you can just type `display` followed by a mathematical expression.
- ▶ For example, to compute $2+2$ type

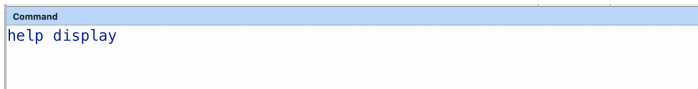
```
display 2+2
```

4

- ▶ The result will be visible in the Results window
- ▶ `display` is called a command in Stata. Stata comes with many commands that will be useful to explore and analyze our data

Help files

- ▶ Commands in Stata come with help files
- ▶ To access a help file type `help display` into the command window



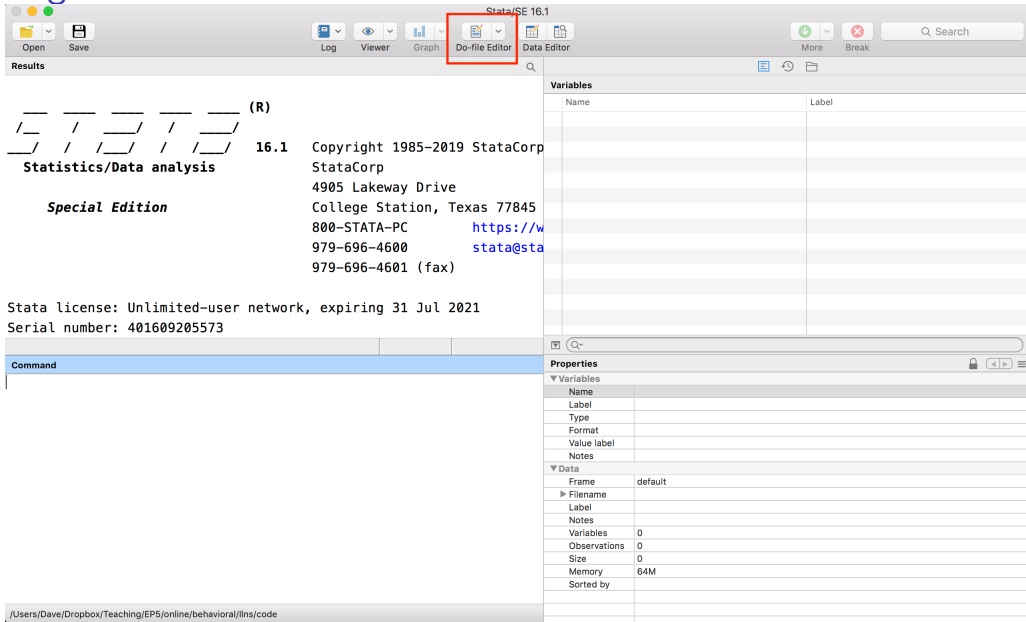
A screenshot of the Stata Command window. The window has a title bar that says "Command". Inside the window, the text "help display" is entered in a blue monospaced font.

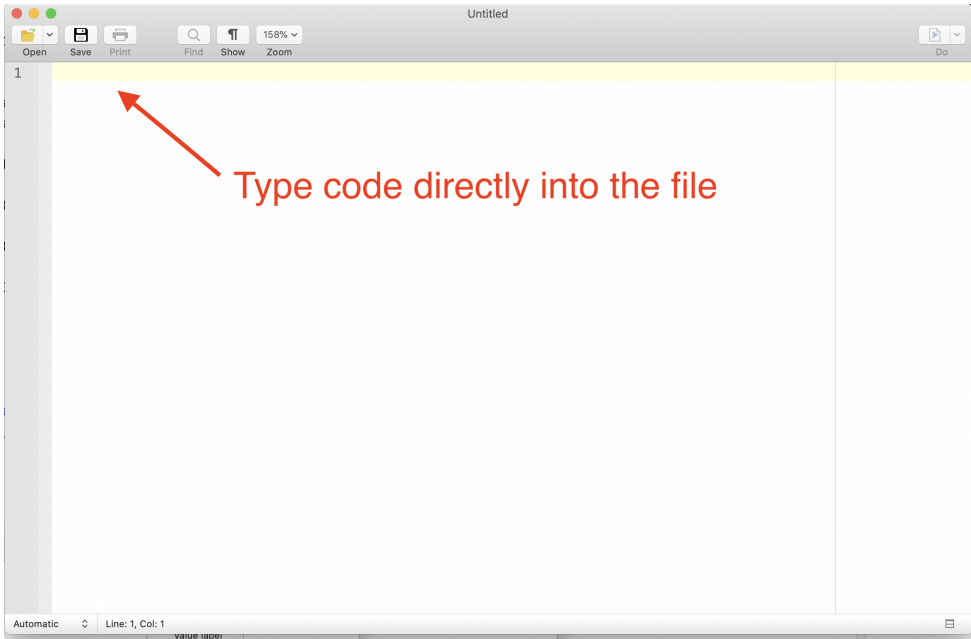
- ▶ This will open up a file that has instructions on how to use the given command

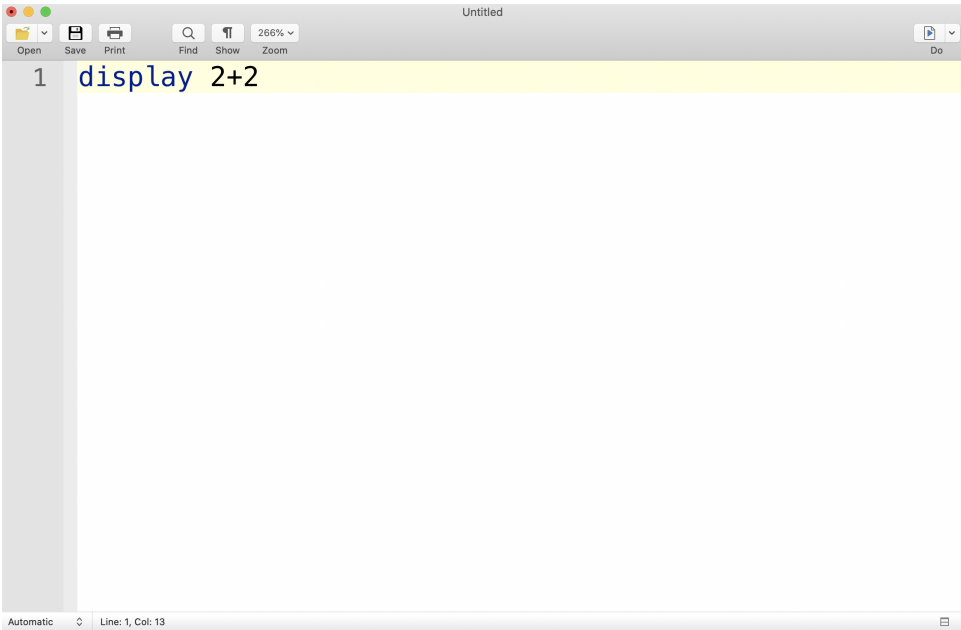
Syntax

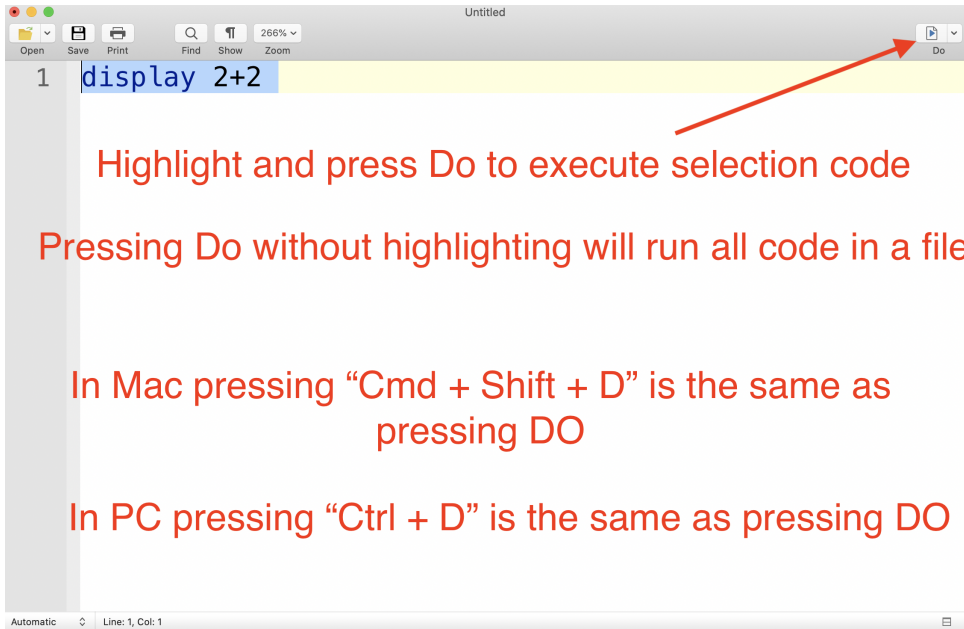
- ▶ Computer languages have different syntax
- ▶ Syntax refers to the format of the code
- ▶ It is similar to the “grammar” of the language
- ▶ If you use incorrect grammar in language, you may not be expressing the right meaning
- ▶ Similarly, if you use the wrong syntax in computer language, you may not get the correct output, and more than likely, an error will be reported

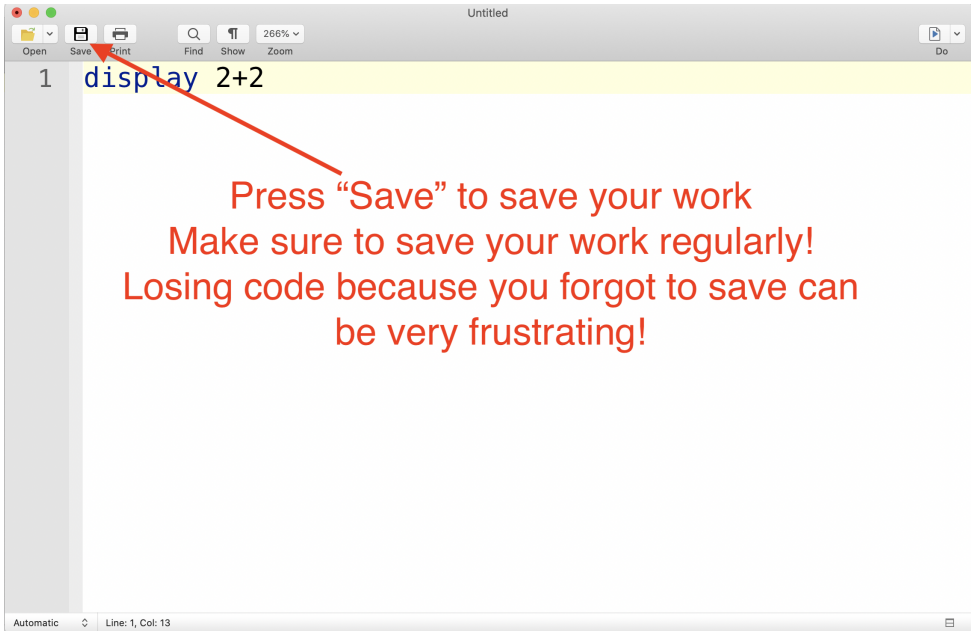
Opening a do-file

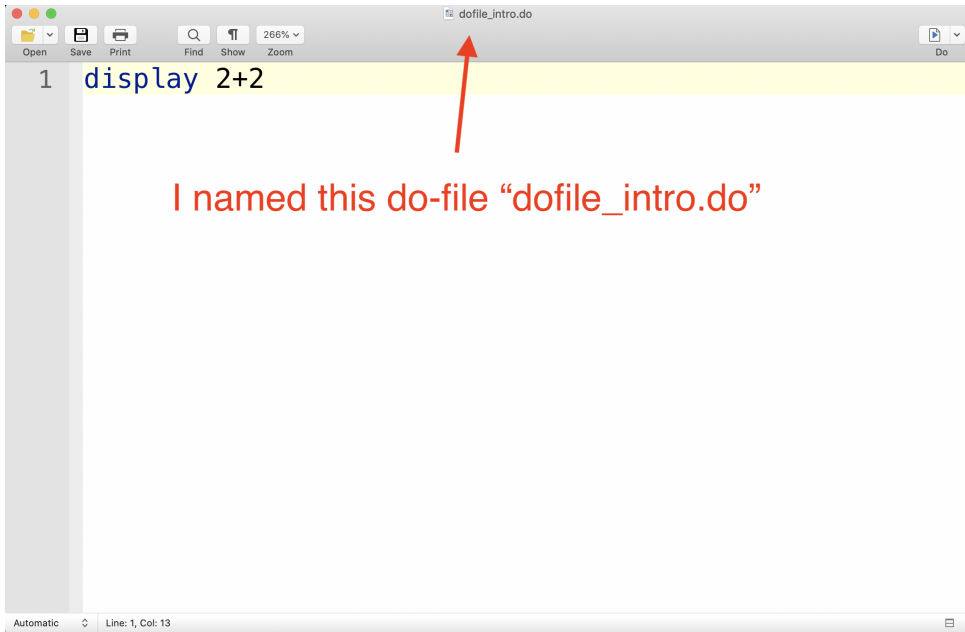


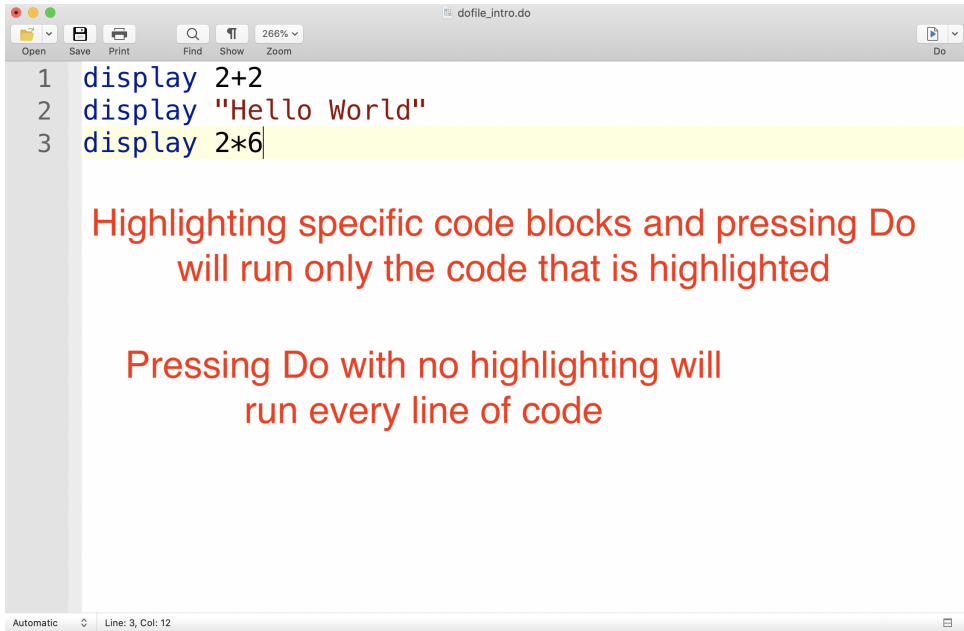


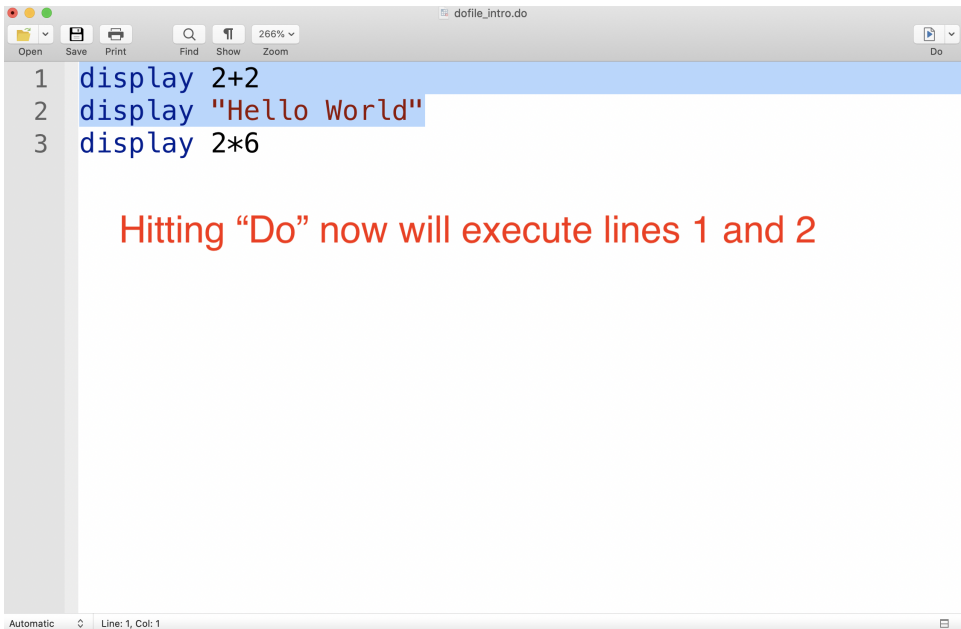












The Working Directory

- ▶ Every folder and file on your computer has an “address” associated with it
 - ▶ Referred to as a **file path**
- ▶ The **working directory** is the the default path used by a program
- ▶ Many commands in Stata (and R) will make use of the working directory

Where to set your working directory?

- ▶ In this class, the working directory should be set to where you save your data after you download it
- ▶ For me, the data is on `Dropbox/Teaching/EP5/online/02_week/videos/data`
- ▶ For you, I suggest making a folder somewhere convenient on your computer named something like EP5, and within that folder creates subfolders for each week of the course
 - ▶ Within that subfolder one of the folders should be named data
 - ▶ Whenever you download data for the week, place it in this folder

Change working directory

- ▶ To change your working directory type “cd” and then the path to folder you want to be your working directory

```
cd "/Users/davidarnold/Dropbox/Teaching/EP5/online/02_week/data"
```

- ▶ This is the path for me, but it will be different depending on where you put the downloaded data for the week

Loading Data

- ▶ To load data into Stata, we utilize the “use” command
- ▶ The general syntax is

```
use filename.dta, clear
```

- ▶ “filename.dta” should be replaced with whatever the name of the file is that you want to load
- ▶ “clear” tells Stata to clear anything that is currently loaded into memory
- ▶ To use this command “filename.dta” must be in the working directory!

```
cd "/Users/davidarnold/Dropbox/Teaching/EP5/online/02_week/data"  
use college_mobility.dta, clear
```

(Preferred Estimates of Access and Mobility Rates by College)

- ▶ Now the college mobility data is loaded into memory